

BIOL-1 Practice Test 06

Module 10: Epigenetics

Instructions: This practice test covers material from Module 10. Answer all questions to the best of your ability.

Part A: Multiple Choice (25 questions)

Choose the best answer for each question.

1. Gene expression means:

- A) Copying DNA
- B) Using a gene to make its protein product
- C) Mutating a gene
- D) Deleting a gene

2. All cells in your body have the same DNA. Different cell types exist because:

- A) They have different DNA
- B) They express different genes
- C) They mutate during development
- D) They lose unnecessary chromosomes

3. A transcription factor is a protein that:

- A) Copies DNA
- B) Translates mRNA
- C) Helps turn a gene on or off
- D) Destroys mRNA

4. A promoter is:

- A) An enzyme that builds proteins
- B) A DNA sequence where RNA polymerase binds to start transcription
- C) A type of RNA
- D) A histone modification

5. An operon is:

- A) A group of genes controlled by one switch
- B) A single gene with many exons
- C) A type of mutation
- D) A eukaryotic structure

6. In the lac operon, when lactose is ABSENT:

- A) The genes are turned on
- B) The repressor binds to the operator and blocks transcription
- C) RNA polymerase makes enzymes
- D) The repressor is inactive

7. In the lac operon, when lactose is PRESENT:

- A) The repressor stays on the operator
- B) The genes remain off
- C) Lactose changes the repressor's shape so it falls off, and genes turn on
- D) The promoter is destroyed

8. The lac operon is an example of:

- A) Constitutive expression
- B) Repressible regulation
- C) Inducible regulation
- D) Epigenetic silencing

9. Epigenetics involves changes in gene expression that:

- A) Change the DNA sequence
- B) Do NOT change the DNA sequence

- C) Only occur in bacteria
- D) Are always permanent

10. DNA methylation typically:

- A) Turns genes on
- B) Turns genes off (silences them)
- C) Has no effect
- D) Deletes genes

11. Histone acetylation:

- A) Tightens DNA around histones
- B) Loosens DNA, making genes accessible (turns them on)
- C) Destroys histones
- D) Blocks translation

12. Euchromatin is:

- A) Tightly packed; genes are off
- B) Loosely packed; genes can be read
- C) Found only in bacteria
- D) A type of mutation

13. Heterochromatin is:

- A) Loosely packed; genes are active
- B) Tightly packed; genes are silenced
- C) Only found during mitosis
- D) The same as euchromatin

14. X-inactivation is:

- A) A mutation on the X chromosome
- B) The random shutdown of one X chromosome in female cells
- C) The deletion of the X chromosome
- D) Only found in males

15. A Barr body is:

- A) An active X chromosome
- B) A condensed, inactivated X chromosome
- C) A type of ribosome
- D) A bacterial structure

16. Calico cats demonstrate X-inactivation because:

- A) They have mutations in coat color genes
- B) Different X chromosomes are active in different patches of skin cells
- C) They are always male
- D) Their genes are not regulated

17. Alternative splicing allows:

- A) One gene to make multiple different proteins
- B) DNA to be replicated faster
- C) Mutations to be corrected
- D) tRNA to carry more amino acids

18. miRNA functions by:

- A) Activating transcription
- B) Blocking translation of specific mRNAs
- C) Replicating DNA
- D) Building ribosomes

19. Which environmental factor can affect gene expression through epigenetics?

- A) Diet
- B) Stress
- C) Toxins
- D) All of the above

20. Identical twins may develop different traits over time because of:

- A) Different DNA sequences
- B) Different epigenetic patterns accumulating over their lifetimes
- C) Different numbers of chromosomes
- D) Mutations in every cell

21. The most common and energy-efficient level of gene regulation is:

- A) Post-translational
- B) Translational
- C) Transcriptional
- D) Chromatin level

22. A housekeeping gene is:

- A) A gene that is only expressed in the liver
- B) A gene that is expressed in all cells at all times
- C) A gene involved in cell division
- D) A gene that is always silenced

23. A tumor suppressor gene that is silenced by heavy methylation (but not mutated) could:

- A) Have no effect on the cell
- B) Contribute to cancer by allowing uncontrolled growth
- C) Make the cell divide more slowly
- D) Turn the cell into a nerve cell

24. The key difference between epigenetics and a mutation is:

- A) Epigenetics changes the DNA sequence; mutations don't
- B) Mutations change the DNA sequence; epigenetics changes expression without changing the sequence
- C) They are the same thing
- D) Epigenetics only affects bacteria

25. An enhancer is a DNA sequence that:

- A) Silences a gene
 - B) Boosts transcription of a gene, even from far away
 - C) Codes for a protein
 - D) Serves as a start codon
-

Part B: Fill in the Blank (10 questions)

- 26. Gene _____ is the process of using a gene to make its protein product.
 - 27. In the lac operon, the _____ protein blocks transcription when lactose is absent.
 - 28. Adding methyl groups to DNA is called _____ and usually silences gene expression.
 - 29. _____ is when acetyl groups are added to histones, loosening DNA and turning genes on.
 - 30. Loosely packed chromatin that allows gene expression is called _____.
 - 31. Tightly packed chromatin that silences genes is called _____.
 - 32. A condensed, inactivated X chromosome is called a _____.
 - 33. _____ allows one gene to produce different proteins by including different exons.
 - 34. Epigenetic changes modify gene expression without changing the _____.
 - 35. The lac operon is an example of _____ regulation because the genes are normally off and are turned on only when lactose is present.
-

Part C: Short Answer (4 questions)

- 36. Explain how the lac operon works. What happens when lactose is absent? What happens when lactose is present?

37. What is the difference between DNA methylation and histone acetylation? Which turns genes on and which turns genes off?

38. Explain how calico cats demonstrate epigenetics (X-inactivation).

39. A heavy smoker's tumor suppressor genes become heavily methylated even though there are no DNA mutations. Explain how this could lead to cancer.